First Year Rossin Engineering
LEHIGH UNIVERSITY

Sensing the World

Reflection Prompt

During this module you will be exploring different concepts and having different experiences. This assignment allows you to reflect on your experience, which reinforces what you learned.

You are expected to type your reflection in essay format. The reflection should start with a personalized summary of what was done that week and how it was accomplished. Then you will describe what you learned and your experience. This should all be from your perspective, so pronouns are expected.

The reflection must be typed and have your name at the top of it and be saved as a PDF. This will be submitted on Gradescope.

Each reflection will be graded as follows:

- Length – is it at least 2 paragraphs – 10%
- Writing – Quality – 20%
- Summary
    - Accuracy – 17.5%
        - Did you describe what was actually done during the specified segment of the module?
    - Completeness -17.5%
        - Did you highlight all the elements from the specified segment of the module?
- Reflection – 35%
    - Completeness and appropriateness

For each criteria considered, with the exception of the length, this is the rubric that will be used for partial credit.

Not done	Beginning	Attempted	Achieved
0% of the Points Required element not attempted.	0% of the Points Substantial errors are made in the materials submitted for this criteria.	50% of the Points Minor errors are made in the materials submitted for this criteria.	100% of the Points Submission is complete and correct for this criteria.

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Here is a sample of a high quality reflection assignment for a multi-week project to consider. It is based on a memory Dr. Winikus had in a lab as a student.

> In this lab we were tasked with using the Buffalo Monitor to program our 8085 chip to control a motor for a conveyor belt. To accomplish this we decided to create a flow chart first to plan out when to read from the sensors and then what to do with the motor. Since real world elements have a lot of variability, we started with testing our calculations and results with a UART interface where we can manually enter values and see how things behaved. Once we were satisfied with the implementation there we integrated our system in and ran through different placement of objects with the sensor to see how the motor responded. Once our response was correct we were able to demonstrate.
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> To be successful with this collaboration we traded off who would do the writing and typing overall. For the flow chart part we leveraged our short hand and both scribbled on the draft to create our plan. When the plan was done my partner cleaned up the drawing in Word while I translated it into comments that would scaffold the code we develop. For the actual code we traded off periodically who would type. In the testing phase we created a bulleted list from discussion to come up with the inputs we needed to test and their expected outputs, I was in charge of writing for that part. When it came time to implement, I took charge of the schematic and then my partner made the physical connections after we discussed it to make sure we were in agreement.
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> Through the collaboration I feel that I was able to improve my programming skills through the discussion with my partner. In the team programming aspect I found it helpful in accomplishing the goals because we had to discuss along the way what we were trying to accomplish. A really nice part of this experience was continuing to work with my partner that we have started to develop a shorthand in communicating, which is fun because all the guys in the class have no idea what we are doing and can't cheat off us. While she was better at programming this assignment then me, like normal, I feel I really was able to contribute to the integration aspect when we were doing the hardware part.
>
> I did find the implementation for the loops and the conditionals challenging, but luckily we did plan carefully and my partner was able to talk through the syntax with me so we were successful. The best part though was adjusting the configuration settings at the end to launch things off the conveyor belt.